SEC P vs NP log 2026-04-29

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-29 02:17 UTC

SEC P vs NP — daily log 2026-04-29

6 cycles recorded

GF(2)-Rank Defect of Term-Indicator Matrix Lower-Bounds Monotone DNF for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid theory — the GF(2) vector matroid M_F realized by the term-indicator matrix $A_F \in F_2^{m\times N}$ of a monotone DNF $F = \vee_{i=1}^m T_i$ (rows $\square\{T_i\}$, columns indexed by variables). Its rank $\rho_2(F) := \mathrm{rank}_{F_2}(A_F)$ is the canonical computable matroid invariant in the Whitney–Welsh tradition; the corresponding 'rank defect' $\Delta(F) := \log_2(m(F)+1) - \log_2(\rho_2(F)+1)$ is a polymatroid-style quantity (subadditive log-rank vs. log-cardinality) that is rarely deployed in monotone circuit lower bounds (Razborov 1985 used Bollobás set-pair / sunflower closure, not GF(2) span of clutter indicators). × Monotone DNF representation size of the k-CLIQUE_n indicator on $N = C(n,2)$ edge variables; equivalently, the minimum-cardinality clutter of minimal terms whose union gives k-CLIQUE_n.
- **Recorded**: 2026-04-29 00:11 UTC
- **Entry ID**: `023dd98c00ee`

Statement

Let F be a monotone DNF on N variables with minimal-term family $H_F = \{T_1,\ldots,T_m\}$, A_F its $m\times N$ indicator matrix, and define $\Delta(F) := \log_2(m+1) - \log_2(\mathrm{rank}_{F_2}(A_F)+1)$. Conjecture: (a) $\Delta(F\wedge G) \le \Delta(F) + \Delta(G) + 1$ and $\Delta(F\vee G) \le \max(\Delta(F),\Delta(G)) + 1$ (approximate submodularity through Boolean operations); (b) $\Delta(F) \le \log_2(N+1)$ for any monotone DNF with $m \le \mathrm{poly}(N)$ (since $\mathrm{rank}_{F_2} \ge 1$ and $m \le \mathrm{poly}$); (c) for the canonical k-CLIQUE_n DNF $F^{n,k}$ with $k = \lfloor n/2\rfloor$, $\Delta(F^{n,k}) \ge n/2 - 3\cdot\log_2 n - 4$ for all $n \ge 6$, witnessing a $\Omega(n)$ gap between log-cardinality and log-GF(2)-rank.

Rationale

The GF(2) span of clique-edge-indicator vectors collapses to a sub-quadratic-rank object (rank $\le C(n,2)$), while the minimal-term count $C(n,\lfloor n/2\rfloor) \approx 2^n/\sqrt{n}$ is exponential — so the LOG ratio is $\Theta(n)$, much larger than any rank deficit attainable by poly-size clutters. Razborov's approximation method tracks subadditive measures across AND/OR gates; an honest submodular Δ that is forced to $\Omega(n)$ on the clique clutter is exactly the missing ingredient for a Razborov-style argument that does not require sunflower closures, since GF(2)-rank is robust under union operations on supports. The bridge from matroid representability to monotone DNF size is concrete (one Gaussian elimination per gate), avoiding the natural-proofs / algebrization barriers that block spectral or Fourier approaches.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2210.12983v2] Poisson multi-Bernoulli mixture filter with general target-generated measurements and arbitrary clutter

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 118, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 78, in run_trial
    F_random = random_poly_dnf(n, n, 2) + random_poly_dnf(n, n**2, 3) + random_poly_dnf(n, n * math.flo
               ^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 56, in random_poly_d
    selected_edges = random.sample(edge_set, w)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence.  "
TypeError: Population must be a sequence.  For dicts or sets, use sorted(d).
```

Judge reasoning

The test crashed with a TypeError in random_poly_dnf (random.sample called on a non-sequence) before any data was produced, so neither the support nor falsification conditions could be evaluated. | next: Fix random_poly_dnf by passing a list (e.g., list(edge_set) or sorted(edge_set)) to random.sample and rerun the 30-seed sweep.

Coarse-Equivalence Invariance of Protocol-Pullback Multiplicity Across Bit-Permuted Gadgets

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) — Coarse Geometry / Roe Algebras (within Geometric Group Theory) × Deterministic 2-party communication complexity, specifically the multiplicity m_Π of the RoeCover extracted from a deterministic protocol's ProtocolPullback for the lifted function f∘G^n
- **Recorded**: 2026-04-29 00:24 UTC
- **Entry ID**: `aa9fe4edbfe5`

Statement

Let G_1 = (X, Y, g, d) be a MetricGadget on small alphabet (|X|,|Y| ≤ 4) and let G_2 be obtained from G_1 by a bi-Lipschitz bijection φ: X×Y → X×Y with distortion ≤ 2 (a coarse equivalence: φ permutes inputs while distorting d by at most a factor 2). Fix any boolean f: {0,1}^n → {0,1} with n ≤ 4. Then for every deterministic protocol Π_1 computing f∘G_1^n, there exists a deterministic protocol Π_2 computing f∘G_2^n with cost cost(Π_2) ≤ cost(Π_1) + O(n), and the multiplicities of their CoarsePullback covers at scale R = diam(G) satisfy m_{Π_2} ≤ 2 · m_{Π_1} + 1. Equivalently, CLC(f,G_1) and CLC(f,G_2) agree up to a constant factor independent of f, instantiating Axiom A1

(Coarse-Lift Functoriality) at the level of the multiplicity invariant that A2 connects to communication complexity.

Rationale

This conjecture directly tests Axiom A1 (Coarse-Lift Functoriality) by asserting that the *multiplicity invariant* m_Π exposed by Axiom A2 (Protocol → Cover) is preserved under coarse equivalences of the underlying gadget. If A1 holds, a bi-Lipschitz φ on X×Y lifts coordinatewise to a bi-Lipschitz map on $(X\times Y)^n$ with the $d_\oplus$ metric (distortion preserved since □^{1} products of bi-Lipschitz maps remain bi-Lipschitz), so any Roe-style cover at scale R for the G_1-lift can be transported to a cover at scale ≤ 2R for the G_2-lift while at most doubling the multiplicity (each transported set may merge with one neighbor at the boundary, contributing the +1 slack). The O(n) cost overhead in Π_2 reflects simulating φ coordinatewise. Verifying this empirically would falsify A1 if multiplicities diverge, and would strengthen confidence that CLC is genuinely a coarse invariant — a prerequisite for using it as a lower-bound technique against communication complexity.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2408.07701v2] A categorical interpretation of Morita equivalence for dynamical von Neumann algebras - [2312.08907v2] Roe algebras of coarse spaces via coarse geometric modules - [1706.00276v1] Counting coarse subsets of a countable group - [2110.02008v2] Lifted Reed-Solomon Codes and Lifted Multiplicity Codes - [1903.06084v8] Lifting coarse homotopies

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 49, in run_trial
    phi_perms = generate_bi_lipschitz_permutations()
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 35, in generate_bi_l
    if is_bi_lipschitz(phi):
       ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 24, in is_bi_lipschi
    if dist(phi[(x1, y1)], phi[(x2, y2)]) > distortion * dist((x1, y1), (x2, y2)):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 18, in dist
    return max(abs(ai - bi) for ai, bi in zip(a, b))
                                          ^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError in the bi-Lipschitz check (dist() called on integers instead of tuples) before any trials produced data, so neither the support nor falsification criteria can be evaluated. | next: Fix the dist() helper to handle the (x,y) pair representation correctly (ensure phi maps tuples to tuples and dist unpacks both arguments), then rerun the 5-seed sweep.

Persistent H_0 Component Count of Sign-Quotient Row Set Caps Real Rank

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Persistent homology / Vietoris-Rips topology of finite metric spaces (Vietoris 1927; Edelsbrunner-Letscher-Zomorodian 2002; Carlsson-Zomorodian 2005; Carlsson 2009). The 0-th persistent Betti number β_0(VR(X,ε)) of a finite metric space (X,d) at scale ε is the number of connected components of the ε-thresholded distance graph, computable by Kruskal/union-find in O(|X|^2 α(|X|)). Rarely deployed in communication complexity (TDA literature is dominated by data-analysis applications; we are aware of <5 papers connecting VR persistence of communication matrices to log-rank / D^cc). × Real rank rk_R(M_f) of the 2^k × 2^k ±1 communication matrix M_f[x,y]=(-1)^{f(x,y)} of a Boolean function f:{0,1}^k × {0,1}^k → {0,1} — the canonical Lovász log-rank / Mehlhorn-Schmidt target for deterministic two-party communication complexity D^cc(f) ≥ log_2 rk_R(M_f).
- **Recorded**: 2026-04-29 00:40 UTC
- **Entry ID**: `bc750201aaf2`

Statement

Let X_f^± ⊂ {±1}$^{\{2}$k}/{±1} be the set of distinct rows of M_f modulo global sign, equipped with the projective Hamming distance d_±(r,s) = min(d_H(r,s), 2^k - d_H(r,s)). Let τ_±(M_f) := β_0(VR(X_f^±, 2^{k-2})), the number of components of the graph linking r,s whenever d_±(r,s) ≤ 2^{k-2}. Then for every Boolean f and every k ∈ {3,4,5,6}, τ_±(M_f) ≤ rk_R(M_f); a single (k,f) with τ_±(M_f) > rk_R(M_f) refutes.

Rationale

Rank-r ±1 matrices have all rows lying in an r-dimensional subspace of R^N, and ±1 vectors in such a subspace are constrained to a structured (centrally-symmetric) finite set whose pairwise distances avoid the 'mid-Hamming' regime around N/2 only by sitting in a small affine pattern; persistence at scale N/4 should therefore not be able to manufacture more than r distinct components after the ±-quotient. This bridges the topological 'shape-counting' of the row metric space to the algebraic rank-bound side of log-rank, a structural inequality complementary to spectral approaches and untouched by natural-proofs (rk_R is the property being lower-bounded, not a hardness predicate on truth tables) and untouched by algebrization (no oracle relativization is used).

Novelty

- Judge: `NOVEL` over 5 arXiv hits

Top hits consulted: - [LIPIcs.ESA.2024.29] A Euclidean Embedding for Computing Persistent Homology with Gaussian Kernels - [s2:609241bc94e187c943bd6d4ec6bcf0c1ababa] Topology in sensor networks - [s2:2505.16879] How high is 'high'? Rethinking the roles of dimensionality in topological data analysis and manifold learning - [PL00009265] Harmonic Analysis, Real Approximation, and the Communication Complexity of Boolean Functions - [s2:f23504fcde864a63a36bc0be33c3a3f97d615] The Nature of Migration and Its Impact on Families in Peru

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7aad0d36.py", line 185, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7aad0d36.py", line 163, in run_trial
    M.append([Fraction(row[i]) for i in range(n)])
```

```
                        ~~~^^^
IndexError: tuple index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any RESULT line, so neither the support nor the falsification condition could be evaluated. Per rule 2, a crashed test yields INCONCLUSIVE. | next: Fix the row-construction bug (tuple index out of range when building M from row entries) and re-run the full 4×30×6 grid to obtain τ_± and rk_R counts.

Secant Rank Lower Bound for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic geometry of secant varieties × Communication complexity
- **Recorded**: 2026-04-29 01:14 UTC
- **Entry ID**: `1a21dd39fb6f`

Statement

The secant rank of the communication matrix for the Disjointness problem DISJ_n is Ω(n), implying a Ω(n) lower bound on randomized communication complexity.

Rationale

Secant rank measures the minimal number of rank-1 tensors needed to approximate a matrix, directly linking to communication complexity via the log-rank conjecture. For DISJ_n, its matrix structure forces high secant rank due to combinatorial constraints, providing a novel algebraic invariant for proving lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.82s`

```
me': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8374.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.5, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.833333333334, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8374.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'con
RESULT: SUPPORTED mean=8373.511111111111 std=0 support_fraction=1.0
```

Judge reasoning

The test used n=6, which is too small to establish Ω(n) bounds. Metric values show no scaling with n, suggesting potential saturation or hidden structure in small cases. | next: Test with larger n (e.g., n=128) to observe scaling behavior of secant rank

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Sign-Rank Distinguishing Tensor for Read-Twice Branching Programs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: discrepancy theory × matrix rigidity
- **Recorded**: 2026-04-29 01:46 UTC
- **Entry ID**: `d2ba6b1a27c2`

Statement

For any read-twice branching program P computing a Boolean function f, the sign-rank of its communication matrix M_P is O(log size(P)), but for the inner product mod 2 function IP_2, the sign-rank of its trivial read-twice BP is Ω(n).

Rationale

This conjecture leverages the connection between sign-rank (a discrepancy-theoretic invariant) and matrix rigidity (a field_B concept). The IP_2 function's exponential lower bound for read-once BPs suggests its read-twice representation must exhibit high sign-rank, while read-twice BPs for general functions should have logarithmic sign-rank due to their structured computation. This bridges the gap between BP complexity and communication complexity barriers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 100, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 77, in run_trial
    A = generate_random_sign_matrix(n, rank)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 47, in generate_rand
    U = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 30, in gaussian_elim
    factor = Fraction(A[j][i], A[i][i])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(0, 0)
```

Judge reasoning

Test crashed with ZeroDivisionError during Gaussian elimination, preventing data collection | next: Debug gaussian_elimination() to handle zero pivots and regenerate matrices

---

SOS Hierarchy Integrality Gap for Random CSPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Sum-of-Squares hierarchy × Constraint Satisfaction Problems (CSPs)
- **Recorded**: 2026-04-29 02:17 UTC
- **Entry ID**: `90b9c49aa839`

Statement

For any constant ε > 0, there exists a CSP instance with n variables and m = Θ(n) constraints such that the SOS hierarchy requires Ω(n^{1-ε}) levels to achieve an integrality gap of 1 - ε for its relaxation.

Rationale

This conjecture builds on the SOS hierarchy's known limitations in handling random CSPs, where the hierarchy's ability to capture exact solutions scales sublinearly with the problem size. It ties to the approach's target of analyzing refutation sizes and integrality gaps, avoiding relativization barriers by focusing on structural properties of random CSP instances.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_566a1d2d.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_566a1d2d.py", line 64, in run_trial
    rank = gaussian_elimination(A)
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_566a1d2d.py", line 29, in gaussian_elim
    factor = Fraction(1, A[i][i])
             ^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(1, 0)
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing data collection. Support condition cannot be evaluated without successful trials. | next: Debug gaussian_elimination() to handle zero pivots, perhaps by adding numerical tolerance or matrix preconditioning